AGENTIC AI · CODING TRACK

Agentic AI Engineer

Design, build, evaluate and deploy AI agents, from LLM basics to multi-agent systems on the cloud.

TRACK	DURATION	FORMAT	COURSE FEE
Coding	**12 weeks, 2 live sessions a week**	**Live online, weekdays, small batch**	**₹25,000 (full fee)**

WHO IT IS FOR
Software engineers, data professionals, tech leads and final-year engineering students.

BEFORE YOU JOIN
Basic programming knowledge helps. Python is taught from the start.

What you will be able to do

- ✔ Choose and work with leading LLMs through their APIs
- ✔ Build agents that plan, use tools, remember and reflect
- ✔ Build stateful workflows with LangGraph and multi-agent teams with CrewAI
- ✔ Write reliable prompts and structured outputs
- ✔ Create RAG pipelines with vector databases
- ✔ Evaluate agents, add guardrails, and deploy on Google Cloud

Course path: 15 modules

1 Module 1: Introduction to AI and Agentic AI

- Evolution of AI; Generative AI vs Agentic AI
- What an AI agent is, and its components
- Single-agent vs multi-agent systems
- Real-world agentic AI applications

Hands-on: Build a simple AI assistant using an LLM

2 Module 2: Python for AI development

- Python basics, functions and OOP
- JSON handling and API calls
- Package management

Hands-on: A Python chatbot that calls an LLM API and parses JSON

3 Module 3: Large Language Models

- Tokens, embeddings and context windows
- GPT, Llama, Gemini, Claude
- Model selection strategies

Hands-on: Work with LLM APIs, run an open-source LLM locally, fine-tune an LLM

4 Module 4: Prompt engineering

- Zero-, one- and few-shot prompting
- Chain of Thought, self-consistency, Tree of Thoughts
- Prompt chaining and structured output

Hands-on: Reusable prompt templates

5 Module 5: AI agent fundamentals

- Agent architecture and lifecycle
- Perception, planning, reasoning, action
- Reflection and memory systems

Hands-on: Build a basic AI agent

6 Module 6: Planning techniques

- ReAct (Reason + Act)
- Plan-and-Execute
- Reflection and self-critique
- Goal decomposition and task breakdown

Hands-on: Build a ReAct agent and a planner agent

7 Module 7: Tool calling, function calling and MCP

- Why agents need tools
- Function and API calling
- Tool selection and routing
- Model Context Protocol basics

Hands-on: Agents with calculator and web search tools; connect an agent to an MCP server

8 Module 8: Retrieval-Augmented Generation (RAG)

- Embeddings and chunking
- Vector and similarity search
- Pinecone and Chroma

Hands-on: A document chatbot

9 Module 9: Memory in Agentic AI

- Short-term and long-term memory
- Memory retrieval
- Vector databases for memory

Hands-on: A memory-enabled agent

10 Module 10: LangChain and LangGraph

- Prompts, chains, output parsers
- LangGraph state, nodes and edges
- Conditional routing, loops, human-in-the-loop, persistence

Hands-on: A stateful agent workflow with LangGraph

11 Module 11: Multi-agent systems

- Collaboration, communication, delegation
- Hierarchies and swarms
- Roles: planner, researcher, coder, reviewer, executor

Hands-on: A multi-agent workflow

12 Module 12: CrewAI framework

- Agents, tasks, crews and tools
- Sequential and hierarchical processes

Hands-on: Research Crew, Resume Builder Crew, Travel Planner Crew

13 Module 13: Agent evaluation

- Agent testing and benchmarks
- Hallucination detection
- Evaluation metrics and reliability testing

Hands-on: Evaluate agent performance

14 Module 14: AI security and guardrails

- Guardrails and prompt injection prevention
- Input, output and tool-call validation
- Content safety and moderation

Hands-on: Implement guardrails for an AI agent

15 Module 15: Deployment

- Agent APIs with FastAPI
- Docker
- Google Cloud Platform

Hands-on: Deploy an agent with Docker on the cloud

CAPSTONE PROJECT OPTIONS

Choose one: Personal AI Assistant, FAQ Bot, Resume Builder (beginner); Document Chatbot, Research Agent, Email Automation Agent (intermediate); Multi-Agent Recruitment System, Financial Analysis Agent, Healthcare Support Agent, Autonomous Research Agent (advanced).

Tools you will use

Python | OpenAI / Gemini / Claude APIs | Llama via Ollama | LangChain | LangGraph | CrewAI | MCP | Pinecone | Chroma | FastAPI | Docker | Google Cloud | GitHub

How every module works

1	2	3	4	5	6
Learn the concept	See a live demo	Practise together	Build a project	Get feedback	Add to portfolio

What is included

- Live interactive classes
- Small batch
- Hands-on project in every module
- Course materials
- Feedback on your work
- Resume and interview guidance
- Certificate of Completion
- Free Peaceful Living course

Reserve your seat

Message me for the next batch dates and class timings.

WhatsApp: +91 85907 90251
bubnasubramanian@gmail.com
Trainer: Bubna Kallidukkil

Certificate of Completion is issued by BK FutureMinds AI after you finish the course, assignments and final project. It is a private training certificate, not a degree or diploma, and not accredited by any university or government body. BK FutureMinds AI is a brand of Bubna Kallidukkil, sole proprietor. The fee is the full fee; no GST is added. Tools and topics may be updated as AI tools change.